

Managing Software in Ubuntu 20.04



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Overview

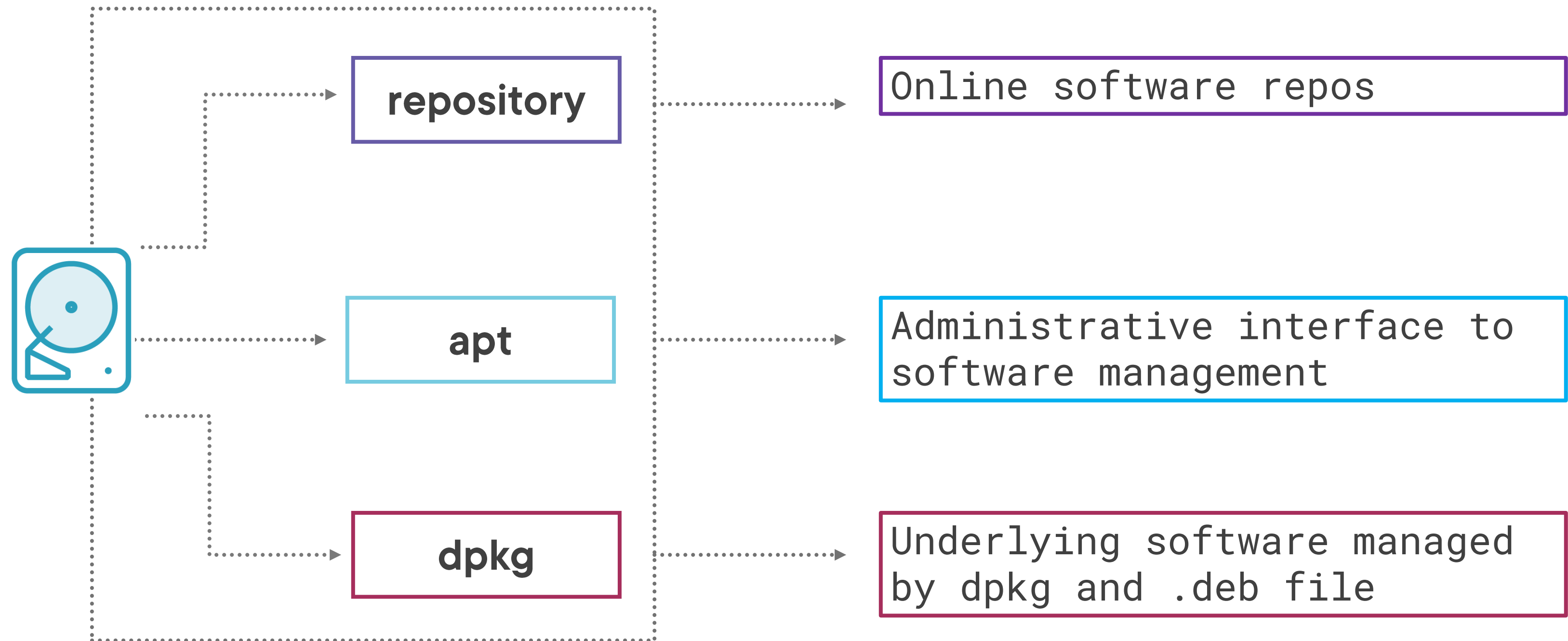


Apt and Dpkg:

- apt software repositories
- sources.list
- dpkg



Software Management



```
$ apt --help  
$ apt moo
```

Super Cow Powers

In Debian based systems the repository package manager is apt and the APT toolset. We can begin by looking at the lighthearted cookie

```
$ sudo apt update
$ sudo apt-get update
$ ssh 192.168.56.102 "sudo apt update"
$ ssh 192.168.56.102 "sudo apt-get update"
```

Apt or Apt-Get

The command `apt` is designed for interactive use and the associated `apt-get` and `apt-cache` toolset for scripts and where there is no terminal attached. Running from a script or remotely via SSH `apt` will give warnings: **WARNING: apt does not have a stable CLI interface. Use with caution in scripts.**

Demo



Let's investigate apt:

- easter eggs
- apt and apt-get
- remote ssh execution



```
$ ls -l /var/cache/apt/pkgcache.bin  
$ sudo apt update  
$ ls -l /var/cache/apt/pkgcache.bin
```

Package Cache

The list of available online packages from the configured APT repositories are stored locally in a cache. If this has been updated recently, we should not need to update the cache before searching or installing software. Checking the last modified time of the cache file will show the last update.



```
$ apt list
$ apt list zz*
$ apt list zzuf
$ apt list --installed
$ apt list --upgradable
$ apt show zzuf
```

Apt-List

Using apt-list we can list all packages, or just installed, upgradeable or available, as well as by pattern. To see information on packages we can read the metadata using apt show.



```
$ apt search apache  
$ apt search --names-only apache  
$ apt search --names-only '^apache'
```

Apt-Search

We can search for packages by name or description. We can also drill down to search just on the name or add regular expressions.



Demo



Working with the APT Cache

- update / last update
- listing packages
- upgrading
- reading package metadata



Demo



Searching and Locating Packages



```
$ sudo apt install -y postfix
```

```
$ sudo apt purge -y postfix
```

```
$ sudo apt autoremove -y
```

Installing and Removing Software

Using apt install we can install a package or list of packages. To delete them we can use remove or purge. The latter will also remove configuration files. Dependency packages are added automatically but they are not automatically removed.



DEBIAN_PRIORITY



low: Trivial questions



medium: Normal questions



high: Only questions that don't have reasonable defaults



critical: Only questions that you must answer



```
$ sudo DEBIAN_PRIORITY=low apt install -y postfix
```

More or Less Questions

By working with the variable, we can control the amount of question asked during the installation.



Demo



Working with Software Installation:

- installing software
- removing software
- working with variables



```
$ sudo apt edit-sources  
$ apt-key list  
$ wget -O- https://apt.releases.hashicorp.com/gpg | sudo apt-key add -  
$ sudo add-apt-repository "deb https://apt.releases.hashicorp.com focal main"  
$ sudo apt edit-sources
```

Repositories

The file `/etc/apt/sources.list` and the extension directory `/etc/apt/sources.list.d/` store the locations of online repositories. We can edit the file directly or add entries to the file using `add-apt-repository`. Packages are digitally signed with a GPG key which will be needed to check the authenticity of packages.



Demo



Managing Repositories:

- adding GPG key
- adding third party repo



```
$ dpkg -l  
$ dpkg -L ufw  
$ dpkg -S /usr/share/ufw/after.rules  
$ sudo dpkg-reconfigure postfix
```

The Core DPKG System

It is the Debian Packager that is used as the basis of APT. We can use this to list packages, package contents and locate a file to its parent package. We can also force installation scripts to re-execute for simple changes to package configuration.



Demo



Working with the Debian Packager:

- dpkg
- dpkg-reconfigure



Summary



Managing Software

- apt or apt-get
- package cache
/var/cache/apt/pkgcache.bin
- searching and listing packages
- installing and removing software
- DEBIAN_PRIORITY=low
- managing keys and repositories
- using dpkg and dpkg-reconfigure



Up Next:

Managing Systemd Services and Targets

